ALI AKARMA

Trustworthy AI & Agentic Systems Researcher
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RESEARCH PROFILE

Undergraduate researcher specializing in trustworthy agentic AI, safe reinforcement learning, adversarial robustness, and AI assurance for autonomous systems. Research experience spans LLM-agent security, multi-agent decision-making, human oversight, verification and validation, and safety evaluation of autonomous AI.

RESEARCH INTERESTS

Trustworthy and safe agentic AI; AI safety and alignment; adversarial robustness and security of autonomous LLM systems; safety-constrained multi-agent reinforcement learning; AI verification, assurance, and human oversight; trustworthy multimodal AI.

EDUCATION

B.S. Information Technology — Islamic University of Madinah, Saudi Arabia *2023 – May 2027*

- GPA: 4.31 / 5.00 — *Merit-based fully funded scholarship (competitive international selection)*
- Relevant coursework: Machine Learning, Algorithms & Data Structures, Probability & Statistics, Database Systems, Software Engineering
- Thesis interest: Trustworthy and verifiable autonomous AI systems under adversarial conditions

RESEARCH EXPERIENCE

Research Intern — AI V&V Lab, KFUPM *Jun 2026 -- Present*

- Developed and evaluated HiVA, a value-aware hierarchical multi-agent RL system for wildfire suppression, using Cell2Fire and testing robustness across 16 distribution-shift stress conditions, including cross-region California↔Saudi transfer, rotated wind, ignitions, and asset layouts.
- Contributed to an agentic extension of vnvspec, a machine-readable V&V framework, by developing governance controls for prompt security, tool authorization, memory leakage, human oversight, and multi-agent coordination, with typed schemas and automated validation.

Undergraduate AI Researcher — Islamic University of Madinah *Jan 2025 -- Present*

- Designed safety-constrained multi-agent reinforcement learning architectures using C-Dec-POMDP and Lagrangian CTDE, achieving 81.5 reward with a 2.3% safety-violation rate across six baselines.
- Analyzed adversarial prompt-injection and jailbreak failure modes in autonomous LLM-agent pipelines; developed LangChain defense middleware with three detection modes and four policy strategies.
- Developed a governance-constrained agentic AI framework incorporating human oversight for safety-critical autonomous decision-making; first author, ICETAS 2026.
- Built 20+ reproducible open-source research systems spanning agentic AI, safety-critical decision-making, AI security, healthcare, and smart infrastructure, with automated testing and archived research artifacts.

RESEARCH PROJECTS & FUNDING

King Salman Center for Disability Research (KSCDR) — *Research Group No. KSRG-2026-321*

- Siddiqui, M. S., Syed, T. A., & **Akarma, A.** (2026). "ADAPT: An Agentic AI Framework for People with Disabilities and Neurodivergence". ***Journal of Disability Research***, *5*(2), e20260830.
  DOI: 10.57197/JDR-2026-0830 | GitHub: github.com/aliakarma/ADAPT
- Syed, T. A., Siddiqui, M. S., **Akarma, A.**, & Formisano, A. (2026). "FedAgent-Chain: A Secure Federated and Agentic AI Framework for Multilingual Disability-Inclusive Employment in AI Cities". ***Smart Cities***, *9*(7), 106.
  DOI: 10.3390/smartcities9070106 | GitHub: github.com/aliakarma/fedagent-chain

SELECTED PUBLICATIONS

Google Scholar: *119 citations | h-index: 8 | i10-index: 5 (October 2026).*

First-Author Papers

- **Akarma, A.,** Syed, T. A., Khan, M., & Ahmad, A. (2026). Privacy Leakage in Federated Learning: Gradient-Based Client Identity Inference and Defenses for Inertial Sensing in Vehicular Edge Networks. **104th VTC Fall Boston** *arXiv:2609.02971*
- **Akarma, A.**, Syed, T.A., Jan, S., Muneer, H., Jilani, A.K. *"Governance-Constrained Agentic AI: Blockchain-Enforced Human Oversight for Safety-Critical Wildfire Monitoring,"* **10th ICETAS 2026**. arXiv:2604.04265
- **Akarma, A.**, Syed, T.A., Jilani, A.K., Jan, S., Muneer, H., Khan, M.A., Yu, C. *"Agents for Agents: An Interrogator-Based Secure Framework for Autonomous Internet of Underwater Things,"* **10th ICETAS 2026**. arXiv:2604.04262
- **Akarma, A.**, Syed, T.A., Hameed, D., Muneer, H. *"Improving Disability Support and Institutional Performance With AI,"* Book Chapter, **IGI Global**. preprints202608.1878

Journal Articles

- Syed, T. A., Akarma, A., Kamal, S., Jan, S., Alkhodre, A. B., & Jamal, A. (2026). Governance-aware autonomous retail coordination in artificial intelligence cities using multi-agent reinforcement learning, blockchain accountability, and federated learning. ***Frontiers in Artificial Intelligence***. 10.3389/frai.2026.1942376 (Co-First Author)
- Syed TA, **Akarma A**, Alatify A, Naqash MT, Alqurashi A. "Agentic AI-enhanced digital twins for Smart City Civil Infrastructure: A Secure, Autonomous and Auditable Management Framework" **PLOS ONE**. 10.1371/journal.pone.0353610 (Co-First Author).
- Jan, S., **Akarma, A**., Syed, T.A., Muhammad, M. A., & Kamal, S. "*EAGF: a four-pillar ethical AI governance framework for trustworthy cybersecurity in 5G renewable energy IoT systems*". ***Scientific Reports*** 2026. 10.1038/s41598-026-63383-5.
- Syed, T. A., **Akarma, A**., Naqash, M. T., Hameed, D., Kamal, S., & Formisano, A. *"Agentic AI for Climate-Resilient Cities: A PRISMA-Guided Review and Digital Twin Framework,"* **MDPI Sustainability**. 10.3390/su18178917
- Syed, T. A., Alshahrani, A., **Akarma, A**., Khan, S., Nauman, M., Lee, I. E., ... & Ullah, A. *"FinNutriAgent (FNA): An Agentic AI for Budget and Nutrition Planning,"* **ETASR**, 2026. 10.48084/etasr.15640

OPEN-SOURCE RESEARCH SOFTWARE

Selected research systems include reproducible experimental pipelines, automated testing, and archived research artifacts.

- **Trustguard** — Constrained multi-agent reinforcement learning framework for trustworthy runtime permission governance using MAPPO and Lagrangian safety constraints. (Repository)
- **Secure Agent Runtime** — Multimodal security framework for autonomous LLM agents, covering prompt injection, provenance, and text, image, audio, and video threats. (Repository)
- **SAS-Shield** — Safety-assurance framework for agentic AI combining conformal testing, formal verification, control-barrier constraints, adversarial evaluation, and cryptographic auditability. (Repository)

Additional research software and reproducibility artifacts: github.com/aliakarma

TECHNICAL SKILLS

AI/ML: PyTorch, Hugging Face Transformers, scikit-learn, NumPy
Agentic AI & LLMs: LLM orchestration, autonomous agents, prompt-injection defense, adversarial robustness
Reinforcement Learning: MAPPO, CTDE, C-Dec-POMDP, Lagrangian constrained optimization
Trustworthy AI: safety evaluation, failure-mode analysis, human oversight, policy-aligned control, auditability
Research Engineering: Python, Git, Linux, Jupyter, pytest, LaTeX, Zenodo

AWARDS & CERTIFICATIONS

- **Merit-Based Fully Funded Scholarship** — Islamic University of Madinah (competitive international selection, 2023)
- **Certificate of Research Excellence** — ICETAS 2026, for research contribution on Agentic AI governance
- **Certificate of Appreciation** — ICBDT 2025, for presented research contributions

LANGUAGES

Languages: Urdu (native), English (professional proficiency), Arabic (professional proficiency)